Biochemistry of Vitamins

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Vitamins characters:

1. They are essential organic compounds for normal health and growth since they are not synthesized in human body.
2. They must be supplied in the diet in the required very small amounts and deficiency of any vitamin in the body results in a disease.
3. They are present in normal food in small amounts.
4. They act as coenzymes.
5. They do not enter in tissue structure as carbohydrates, fats and proteins.
6. They are not oxidized for energy production as carbohydrates, fats and proteins.

Classification:

- Vitamins can be classified according to their solubility into two main groups:

I- Fat-soluble vitamins. **II- Water-soluble vitamins.**

Fat-soluble vitamins

Vitamin A (retinol)

(Anti-night blindness, Anti-xerophthalmia vitamin)

Chemistry:

- The active forms of vitamin A are retinol, retinal (retinaldehyde), and retinoic acid.

- It is available to the human either in the form of retinol or as a precursor (or provitamin) that is called **carotenoids** or carotenes (hydrocarbon pigments, yellow or red).

Sources:

- **Plant origin**: All pigmented (particularly yellow) vegetables and fruits and the leafy green vegetables supply provitamin A (carotenes). It is present in carrots, tomatoes and potato.

- **Animal origin**: It is present in meat, liver, milk, butter and egg yolk supply retinol palmitate.
- In fish liver oils retinol is present as esters of fatty acids, chiefly stearic, palmitic and higher unsaturated fatty acids.

Functions of vitamin A:

I- Role of vitamin A in vision (rhodopsin cycle):

- The light receptors of the eye are the rod and cone cells in the retina. **Rods** are responsible for vision in dim light. They contain a photosensitive pigment called **rhodopsin** (or visual purple) which is formed from **11-cis retinal** and **opsin** (a simple protein).

Cones Rods are responsible for vision in bright and colored light and contain a photosensitive pigment called **iodopsin**. It is formed from retinal and a protein called **photopsin**.

II. Role of retinoic acid (hormone A):

1- glycoprotein synthesis

2- transferrin.

3- growth of bones and teeth.

4 - anticancer agent

5- It is immunostimulant

III. Antioxidant activity of retinoids:

Deficiency

1. **In the eye:** Night blindness
2. **Growth retardation**
3. **Increased liability to cancer and anemia**.
4. **The skin** becomes rough and hyper-keratinized (goose skin, xeroderma).
5. Roughness of mucous membrane of the urinary tract, genital tract, gastrointestinal tract and the respiratory tract leading to repeated infection in these sites.

Vitamin D (calciferol)

- There are two types of vitamin D : vitamin D_2 derived from ergosterol present in plants , and vit-D_3 derived from cholesterol .

Sources:

I- Provitamins:

- 7-dehydrocholesterol in animals and man only. It is formed in intestinal mucosa from cholesterol then passes to the skin, whereby ultraviolet rays of the sun convert it into cholecalciferol (Vitamin D_3). This demonstrates the importance of outdoor exposure of infants to sunlight.

II- Vitamin D:

- The richest source of vitamin D_3 is fish liver oils, e.g., cod and shark liver oil.

- Egg yolk, liver, butter and milk are a poor source of vitamin D_3.

*** Activation of vitamin D (synthesis of calcitriol)**

- **In the liver** vitamin D_3 is hydroxylated by 25-hydroxylase to 25-(OH)-D_3 that pass to circulation and is the major form of vitamin D in the circulation and the major storage form in the liver.

- **In the proximal renal tubules**, (to less extent in **bone and placenta)**, the 25-(OH)-D_3 is further hydroxylated to 1,25-dihydroxy-D_3 (**Calcitriol**) by 1-hydroxylase enzyme .

Functions of active vitamin D

Vitamin D functions as a hormone more than as a vitamin. The target tissues include the **Intestine, bone** and **kidney.**

1- In the intestine

1, 25-(OH)$_2$-D$_3$ induces synthesis of a protein required for calcium absorption (Ca^{2+}-binding protein). 1,25-(OH)$_2$-D_3 also increases phosphate absorption from intestine. This increases the plasma level of Ca^{2+} and phosphate.

2- In the bone,

I) 1,25-(OH)$_2$-D_3 and PTH act synergistically to promote mobilization of calcium and phosphate from bone (deminerilization) .

II) 1,25-(OH)$_2$-D_3 increases the synthesis of osteocalcin (Ca^{2+}-binding protein) of bone, thereby influencing the mineralization of bone tissues.

3- In the kidney, 1,25-(OH)$_2$-D_3 enhances the reabsorption of Ca^{2+} and phosphate.

Deficiency:

- It causes rickets in young children and **osteomalacia** in adults

* **The causes:**

- Decrease intake of vitamin D
- Minimal exposure to sunlight.
- Diseases causing fat malabsorption.
- Severe liver and kidney diseases due to failure of conversion of 25-(OH)- D_3 to 1,25-$(OH)_2$-D_3.

Vitamin D toxicity (Hypervitaminosis):

Usually occurs due to self medication. It results in hypercalcaemia that leads to abnormal calcification in soft tissues like, Kidney leading to kidney stones, stomach causing nausea and vomiting and blood vessels leading to hypertension.

Requirements: Infants 400 IU/day. Pregnant and lactating females 800 IU/day.

Vitamin E (tocopherols)

(Anti-sterility vitamin)

Chemistry:

- Vitamin E or tocopherols. Tocopherols are organic compounds.

Sources:

- Vegetable oils, e.g.,

Functions:

Vitamin E plays an important role as **antioxidant**.

1. Vitamin E appears to be the first line of defense against lipid peroxidation.
2. Vitamin E appears to play a role in cellular respiration, by stabilizing coenzyme Q (prevents its oxidation).
3. It prevents hemolytic anemia.
4. It has antisterility effect especially in animals.

*** Manifestations:**

1- Anaemia which may be due to decreased production of hemoglobin or hemolysis of RBCs .

2- Sterility in rats .

3- Central nervous system changes .

4- Cancer stomach .

Vitamin K

(Anti-haemorrhagic vitamin)

Chemistry:

Two types of vitamin K are existed:

1- Naturally occurring vitamin K (fat soluble):

Vitamin K_1 (phylloquinone), is the major form of vitamin K in plants.

Vitamin K_2 (menaquinone), is formed inside the intestine by intestinal bacteria.

2- Synthetic (water soluble):

Vitamin K_3 (Menadione) is water soluble and it is most potent member .

Sources:

-Vitamin K_1 is present in spinach, cauliflower and cabbage.

- The main source of vitamin K2 is intesinal bacteria.

Functions:

I- Vitamin K is antihemorrhagic

A) Vitamin K is essential for synthesis and activation of blood clotting factors,.

II- Activation of osteocalcin in bones (calcium-binding protein). Calcitriol stimulates the synthesis of osteocalcin . Vit-K activates osteocalcin.

Deficiency :

*** The Causes:**

1. Deficient intestinal bacteria (sterilization of large intestine) e.g. long use of antibiotic and in newly born infants.
2. Fat malabsorption syndromes associated with pancreatic dysfunction, biliary disease, intestinal mucosal atrophy or any other cause of steatorrhea.
3. Long use of dicumarol. This anticoagulant drug antagonizes vitamin K (by competitive inhibition) and leads to the decrease of prothrombin activation.
4. Liver diseases.

*** Effects:**

- Hemorrhagic manifestation in skin and mucous membranes due to defect in blood coagulation mechanism.
- Prolonged clotting time .

N.B., Deficiency of vitamin K does not occur as a result of defecient intake because it is widely distributed in foods and it is also synthesized by bacteria in large intestine.

II- Water-soluble vitamins

This group includes:

1. Vitamin C. **2. Vitamin B Complex**

Vitamin C or “L-Ascorbic Acid” (Antiscorbutic vitamin)

Sources:

- Fruits especially citrus fruits (lemon, orange) melon and strawberry. Gawafa is very rich in vitamin C.
- Vegetables especially green leafy vegetables such as lettuce, tomatoes, raw cabbage, green peppers and germinating seeds are rich source of the vitamin.
- Liver and adrenals are animal sources of vitamin C are useless due to destruction of vitamin C by coocking.
- Milk is a poor source of vitamin C.

Physiological functions:

1. Formation of collagen:
- Collagen is essential for the **integrity of connective tissue.**
- Vitamin C is also necessary for **bone formation.**
- vitamin C accelerates the **healing of wounds and fractures of bones.**
- Collagen is a component of the **capillary walls**, so vitamin C deficiency is associated with capillary fragility.
2. Vitamin C influences the biosynthesis of adrenal cortical hormones and catecholamine (adrenaline and nor adrenaline) as it is present in high concentration in the adrenal gland, especially in periods of stress.
3. Ascorbic acid is required for the formation of carnitine by hydroxylation of procarnitine. Carnitine stimulates fatty acid oxidation in mitochondria. Decreased level of vitamin C may lead to lowered energy levels and muscle weakness.
4. Ascorbic acid is needed for bile acid formation from cholesterol.
5. It aids in the absorption of iron by reducing it from the ferric (Fe^{3+}) to ferrous (Fe^{2+}) state neccessery for absorption.

6. Ascorbic acid is necessary for the conversion of folic acid into its physiologically active form tetrahydrofolic acid.
7. Ascorbic acid acts as a general water-soluble antioxidant. It spares vit. A, E and some B vitamins by protecting them from oxidation.
8. In large doses, vitamin C may inhibit the formation of carcinogenic N-nitrosocompound during cooking and digestion and lowered the risk for gastric and esophogeal cancers.

The use of vitamin C in large doses ameliorates the symptoms of the common cold and increases the resistance of the body.

Deficiency:

Causes of deficiency:

1. in chronic disease such as cancer, alcoholism, cigarette smokers and users of contraceptive pills.

2. The classic deficiency disease **Scurvy** occurs only after chronic deprivation of vitamin C.

The clinical manifestations of scurvy include:

1. Easy bruising and haemorrhages under the skin due to increased capillary fragility.
2. Gums become swollen, tender and bleed easily.
3. The teeth become loose and may fall.
4. Delayed wound healing, osteoporosis and anemia.
5. Decreased immunocompetence.

Vitamin B Complex

- Vitamin B complex includes the following members:

- **Thiamin (vit B_1).** - **Riboflavin (Vitamin B_2).**
- **Niacin (P.P.F.).** - **Pyridoxine (Vitamin B_6.)**
- **Pantothenic acid.** - **Biotin (Vitamin H.).**
- **Folic acid.** - **Vitamin B_{12}.**
- **Lipoic acid** - **Inositol.**
- **Choline.** **-P-amino benzoic acid (PABA).**

General characters of vitamin B complex:

- All are soluble in water.
- All are present in the same sources (in whole grains, cereals, liver, and yeast).
- All members of vitamin B complex serve as coenzymes or cofactors in enzymatic reactions.
- All members of vitamin B complex are absorbed in the intestine and transported in the portal circulation.
- The tissue stores of most members of vitamin B complex are minimal.
- Toxic effects are relatively uncommon.
- They are heat stable except vitamin B1.
- They destroyed by light except niacin.
- They are stable in acid medium but unstable in alkaline medium.

Thiamin
(Vitamin B_1 or Antiberiberi)

Sources:

- **Plants:** Seeds as peas, beans, whole cereal grains, bran and yeast.
- **Animals:** liver, eggs and milk.

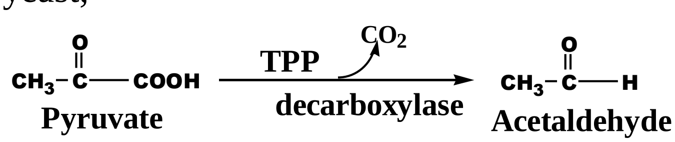
Absorption and Metabolism:

- Thiamin is absorbed in the intestine and transported in the portal circulation to the liver and most tissues.
- Thiamin pyrophosphokinase present in the liver, erythrocytes and cells of cerebral cortex converts thiamin into thiamin pyrophosphate coenzyme form.

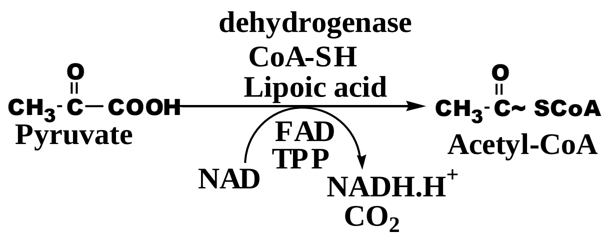
Functions:

- Its coenzyme, TPP (codecarboxylase) is necessary for decarboxylation of α-keto acids which includes:

1. Simple decarboxylation which needs TPP only as a coenzyme. This reaction occurs in yeast,



2. Oxidative decarboxylation



3.transketolase is the enzyme

4. It is necessary for optimal growth of infants and children.

5. It increases the activity of acetyl choline at nerve endings by inhibiting acetyl choline esterase enzyme.

Deficiency:

- **Beriberi** is a nutrition deficiency disease caused by carbohydrate-rich thiamin-low diets, i.e., polished rice or highly refined foods such as sugar and white flour. Certain raw fishes contain a heat labile enzyme **thiaminase** that destroys thiamin, but this is not considered to be critical in human nutrition.

- Deficiency of vitamin B_1 and TPP leads to:

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Manifestations:

Beriberi: - There are two forms of beriberi:

I. Wet beriberi: Affects mainly the cardiovascular system and it is characterized by extensive edema and high output cardiac failure.

II. **Dry beriberi:** Affects mainly the nervous system and it is associated with: Polyneuritis (Inflammation of peripheral nerves), hyperesthesia, and muscle wasting and loss of weight.

Riboflavin (Vitamin B_2)

Absorption and Metabolism:

- In the intestinal mucosal cells, flavin mononucleotide (FMN) is formed by ATP-dependent phosphorylation of riboflavin catalyzed by flavokinase enzyme. This enzyme is competitively inhibited by chloropromazine drug. Whereas, FAD (the predominant tissue form) is synthesized by a further reaction with ATP in which the AMP moiety of ATP is transferred to FMN.
- Both flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN) are active forms of vitamin B_2. They are absorbed in the intestine and bind to albumin for their transportation in the blood.

Functions:

- Riboflavin is converted to FMN and FAD coenzymes which act as hydrogen carriers for certain enzymes called flavoproteins.
- Flavoprotein enzymes are wide spread and involved in oxidation-reduction
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Deficiency:

1. In the mouth: Red lips and shiny-Angular stomatitis (inflammation of angles of mouth). Glossitis (inflammation of tongue).
2. In the skin: Seborrheic dermatitis (inflammation of sebaceous glands of skin).
3. Occular disturbance: Photophobia and viscularization of cornea.
4. Synthesis of proteins is impaired in severe riboflavin deficiency.

Requirements:

Niacin (Nicotinic acid or Pellagra Preventing Factor, PPF)

Sources:

- Niacin can be made endogenously from the amino acid tryptophan. Each 60 mg trytpophan can be converted to one mg niacin. This conversion requires vitamin B_6, thiamin and riboflavin as coenzymes.

- Whole grain cereals, yeast, milk, leafy green vegetables and meat. Meat is rich in tryptophan, so it is important source of niacin. Corn is poor in both niacin and tryptophan. It contains niacin in the form of nicotinyl esters. The body cannot hydrolyze these esters during digestion.

Functions:

- Nicotinic acid in the form of nicotinamide enters in the formation of:

1. Coenzyme I (NAD): nicotinamide adenine dinucleotide
2. Coenzyme II (NADP): nicotinamide adenine dinucleotide phosphate.

- These two coenzymes function as hydrogen carrier and they are essential for many biochemical oxidation-reduction reactions. These reactions are important in carbohydrate, protein and lipid metabolism.

3. Coenzyme III (NMN): nicotinamide mononucleolide which acts as a coenzyme for synthesis of taurine from cysteine which is used for the synthesis of bile acid (taurocholic acid).

- Nicotinic acid and its amide are stimulant to C.N.S. Nicotinic acid is a vasodilator. Nicotinic acid only has been used therapeutically for lowering plasma cholesterol.

Deficiency:

- Deficiency of niacin interferes with biologic oxidation of carbohydrate, lipids and protein metabolism. It leads to a disease called **Pellagra**.

Causes of deficiency:

- Pellagra is usually associated with deficiency of niacin, tryptophan or pyridoxine.
- Corn is deficient in both niacin and tryptophan. Thus, people who depend on corn as a major source of protein as some farmers develop pellagra.
- In some diseases of tryptophan metabolism as: Hartnup disease and in carcinoid syndrome or after isoniazide therapy for T.B., pellagra can occur.

Pellagra is characterized by:-

1. Dermatitis: The exposed skin becomes dry, rough and scaly with brown discoloration, glossitis and stomatitis are seen.
2. Diarrhea.
3. Dementia.